

Request for Proposal

Software Development & QA Services

Technical Overview – Quality / Testing

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Software Quality Assurance & Testing Services RFP

1. Purpose of This RFP

Assa Abloy America Residential is issuing this Request for Proposal (RFP) to identify and engage a qualified **Software Quality Assurance (SQA) services vendor** for a **long-term strategic partnership**. The selected vendor will augment our internal Quality Engineering organization while assuming **full ownership of testing activities** for our **consumer IoT electronic locks ecosystem**.

We are seeking a **mature, process-driven QA partner** capable of delivering consistent quality outcomes across hardware, firmware, mobile, cloud, and web platforms.

2. Product & Business Context

Our products are **Consumer Electronic locks**, combining physical hardware with a robust digital ecosystem. The platform includes:

- Physical lock devices
- Firmware Testing
- Mobile applications (iOS and Android)
- Cloud backend services hosted on AWS
- APIs Test
- Web portals for users and operations

Given the security-sensitive nature of IoT products, quality, reliability, and performance are **critical business enablers**.

3. Engagement Model & Expectations

The engagement model combines **staff augmentation with full test ownership**, enabling vendors to operate as an extension of the internal QA organization while fully owning execution, planning, and quality visibility.

Key expectations include:

- Long-term engagement under a **fixed-price commercial model**

- Offshore delivery with required overlap during **PST mornings hours for calls**
- Strong collaboration within agile development processes

Final release authority (go/no-go) remains with Assa Abloy while vendors are expected to provide data-driven quality recommendations and risk assessments.

4. Scope of Testing Services

Vendors must demonstrate **proven experience across all mandatory areas**, including:

- Test Env Setup
- IoT device & Firmware testing
 - FW releases cycle testing for all basic customer operations and cases where system can break.
 - FW compatibility testing on 3 older apps and cloud
 - OTA from manufacturing and OTA bundles
- Hardware, mainboard, battery, power testing, Current tests, Manufacturing testing, reliability test, RFID test
- Mobile application testing (iOS / Android) all brands
 - For App releases testing on all supported locks and devices complete test suite including positive, negative and corner cases. (Current test case count For Halos: 8500+ which we are running on IOS, Android, on different lock types: Aura BT, Halo touchscreen, Halo keypad, Halo select, Halo select plus, FP, Puck, spark, Aura reach) For Unite we run using multiple users (ex: PMA, PMU,IA, IU, Tenant, Guest) all supported apps.
 - Compatibility testing on different IOS and Android devices screen size and supported O/S
 - Testing using multiple routers for compatibility and border routers and devices based on the lock type WIFI/Matter etc. for product launch.
- Web portal testing
- API testing
- AWS / cloud testing

- DR testing, partner smart home integration testing
- Backend testing
- Compatibility of 3 older all brands' apps
- Automation testing (UI, API, backend, cloud)
- Performance and load testing

The vendor will own **test strategy, planning, execution, automation development, defect management, and reporting.**

5. Quality Processes & Deliverables

Vendors are expected to operate within an agile framework and actively participate in:

- Daily standups
- Sprint planning and demos
- Retrospectives
- Defect triage meetings

Required deliverables include test strategies, test plans, test cases, automation frameworks, traceability matrices, and release sign-off reports SWQA should follow project plan and **stick to timelines and deliver on time. Jira** is the standard tool for defect and test tracking.

6. Production & Post-Release Support

Production validation is a core responsibility:

- Release validation begins on **release day**
 - Core validation completed within **one day**
 - App releases are phased production rollout support may continue for **up to one week**
 - Work with customer care and marketing sales team on any field issues to recreate, verify the fix per requirements and production deployment.
 - Long term testing to collect data for every production release.
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7. Automation & Performance Focus

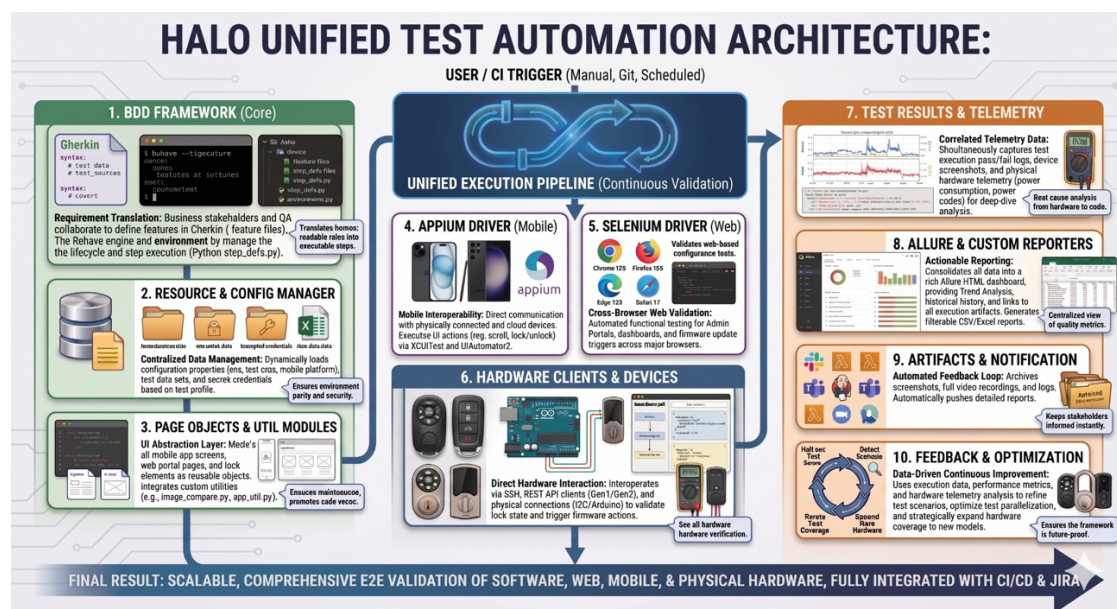
- Vendors will **own and maintain automation frameworks**
- Automation execution is required for **weekly builds**
- Performance and regression testing is required for **every regression build**
- Automation and performance results must be clearly reported and trended

Lock Independent cases and without HIL cases:





Lock Dependent cases on HIL



Technologies

Software: J-Flash lite, SN-Link ICP (SONIX), Joulescope, Keysight 14548, Keysight Command Expert, Segger J-Link, nRF connect for Desktop, A2 Matter Provisioning Tool, NXP provisioning tool, External Provisioning controller, Aura EOL tool, Efinger, Simplicity Studios v4 and v5, VulcanJtag , Simplicity Comander, Uniflash, Z-wave PC controller, Salee Logic Analyzer, Wireshark, Sublime Text, CBA software, Vysor, Android Studios, and VS code, Zwave sniffer, FET-Pro430-Lite,

Hardware: Keysight N6705B, Kpack, CBA V, J-link Segger, Uart Provisioning tool, Salee, Zwave Controller, Zwave sniffer, Nordic BLE dongle, For Automation: Arduino, Customer made Hardware fixtures, Raspberry Pi controller

Automation/Performance test tools: Appium, WebdriverIO (test runner for mobile), Appium MCP (mobile automation via AI), iOS (XCUITest driver), Android (UiAutomator2 driver), Appium-Python-Client , Selenium, Behave , Allure (allure-behave, allure-python-commons), Playwright, Playwright MCP, Node.js, JavaScript , Java, Python, TypeScript, Test NG, Cucumber, GraphQL, Git, Bitbucket, Visual Studio Code, GitHub Copilot (AI-assisted development), Xcode (iOS development/debugging), Android Studio (Android development/debugging) , HIL (Hardware-in-the-Loop) system, JMeter, Browser stack.